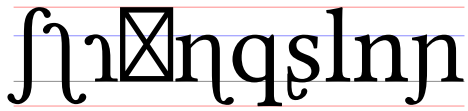
Test sheet: ʃ U+0283, ɻ U+0285, l, ɲ

To-do item (2026-11-07): “semibold ɻ has exactly the bounding box of ʃ; semibold and semibold italic l below are regular’s $(135, -110)$; italic ɲ above is at $Y = 654$, raised by stage 8 because its arch is 9 taller than n’s.”

1. Shapes

Each style: original font, then patched (stage 2 redraws ɻ and adds U+E27F, ɹ with descender, which the original lacks). Lines: ascender 698, x-height (429 / 434), baseline, descender −232. Glyphs: ʃ ɻ ɹ U+E27F ɳ q ʂ l n ɲ. The last row is Gentium Plus (2048 upm, scaled to the same body size; its lines are not drawn).

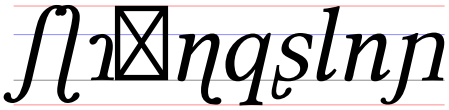
Regular



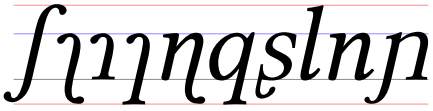
patched



Italic



patched



Semibold



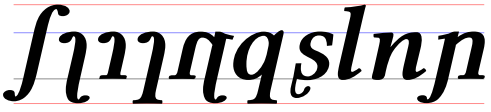
patched



Semibold italic



patched



Gentium Plus



	ɻ original bbox, adv	ɻ patched bbox, adv	above	below	U+E27F bbox, adv	above	below
regular	$(-33, -214, 374, 550)$, 344	$(15, -221, 388, 442)$, 341	$(170, 645)$	$(202, -306)$	$(15, -234, 334, 442)$, 348	$(168, 645)$	$(220, -319)$
italic	$(83, -214, 313, 708)$, 304	$(100, -229, 354, 435)$, 312	$(307, 645)$	$(88, -319)$	$(63, -238, 320, 435)$, 340	$(305, 645)$	$(145, -319)$
semibold	$(-71, -211, 376, 698)$, 328	$(7, -229, 422, 445)$, 367	$(182, 645)$	$(214, -319)$	$(7, -234, 355, 445)$, 372	$(182, 645)$	$(231, -319)$
semibold italic	$(73, -211, 344, 698)$, 328	$(93, -229, 382, 445)$, 368	$(167, 645)$	$(131, -319)$	$(46, -234, 354, 445)$, 372	$(163, 645)$	$(170, -319)$

ɻ (squat reversed esh) is now ɹ’s fishhook and stem with ɳ’s retroflex hook: x-height top, hook down to −221 (regular) or −229, short of the descender, like the other retroflex hooks. U+E27F is ɹ’s fishhook and stem run down into q’s descender serif.

2. In use (patched fonts, no CGJ)

Esh in transcription: English, syllabic and diacritic-marked fricatives.

Regular [ʃɪp] [fɪʃ] [tʃɜːtʃ] [ʃʷuːt] [ʃ̩ː] [pʃ̩t] [ʃ̻] [ʃ̰] [ʃ̲] [ʃ̟] [ʃ̊] [ʃ̃]

Italic *[ʃɪp] [fɪʃ] [tʃɜːtʃ] [ʃʷuːt] [ʃ̩ː] [pʃ̩t] [ʃ̻] [ʃ̰] [ʃ̲] [ʃ̟] [ʃ̊] [ʃ̃]*

Semibold **[ʃɪp] [fɪʃ] [tʃɜːtʃ] [ʃʷuːt] [ʃ̩ː] [pʃ̩t] [ʃ̻] [ʃ̰] [ʃ̲] [ʃ̟] [ʃ̊] [ʃ̃]**

Semibold italic ***[ʃɪp] [fɪʃ] [tʃɜːtʃ] [ʃʷuːt] [ʃ̩ː] [pʃ̩t] [ʃ̻] [ʃ̰] [ʃ̲] [ʃ̟] [ʃ̊] [ʃ̃]***

Regular [ʃ̌] [ʃʲ] [ʃ̯] [ʃ̩]

Italic *[ʃ̌] [ʃʲ] [ʃ̯] [ʃ̩]*

Semibold **[ᷓ] [ᶢ] [ᷔ] [ᷕ]**

Semibold italic ***[ᷓ] [ᶢ] [ᷔ] [ᷕ]***

Squat reversed esh (Sinological apical vowels): next to ɿ and with tone marks.

Regular shī [ʂɿ^{55}] shì [ʂɿ^{51}] rì [ʐɿ^{51}] zhī [tʂɿ^{55}] chī [tʂ^{h}ɿ^{55}] sī [sɿ^{55}] cí [ts^{h}ɿ^{35}]

Italic *shī [ʂɿ^{55}] shì [ʂɿ^{51}] rì [ʐɿ^{51}] zhī [tʂɿ^{55}] chī [tʂ^{h}ɿ^{55}] sī [sɿ^{55}] cí [ts^{h}ɿ^{35}]*

Semibold **shī [ʂɿ^{55}] shì [ʂɿ^{51}] rì [ʐɿ^{51}] zhī [tʂɿ^{55}] chī [tʂ^{h}ɿ^{55}] sī [sɿ^{55}] cí [ts^{h}ɿ^{35}]**

Semibold italic ***shī [ʂɿ^{55}] shì [ʂɿ^{51}] rì [ʐɿ^{51}] zhī [tʂɿ^{55}] chī [tʂ^{h}ɿ^{55}] sī [sɿ^{55}] cí [ts^{h}ɿ^{35}]***

Regular [ʂɿ̄ ʂɿ́ ʂɿ̌ ʂɿ̀] [sɿ̄ sɿ́ sɿ̌ sɿ̀] [ʂɻ̃] [ʂɻ̃] [ᷓ ᷔ]

Italic *[ʂɿ̄ ʂɿ́ ʂɿ̌ ʂɿ̀] [sɿ̄ sɿ́ sɿ̌ sɿ̀] [ʂɻ̃] [ʂɻ̃] [ᷓ ᷔ]*

Semibold **[ʂɿ̄ ʂɿ́ ʂɿ̌ ʂɿ̀] [sɿ̄ sɿ́ sɿ̌ sɿ̀] [ʂɻ̃] [ʂɻ̃] [ᷓ ᷔ]**

Semibold italic ***[ʂɿ̄ ʂɿ́ ʂɿ̌ ʂɿ̀] [sɿ̄ sɿ́ sɿ̌ sɿ̀] [ʂɻ̃] [ʂɻ̃] [ᷓ ᷔ]***

Regular sī [sɿ^{55}] cí [ts^{h}ɿ^{35}] [sɿ̄ sɿ́ sɿ̌ sɿ̀] [sɻ̃] [ʂɿ sɪ sɻ]

Italic *sī [sɿ^{55}] cí [ts^{h}ɿ^{35}] [sɿ̄ sɿ́ sɿ̌ sɿ̀] [sɻ̃] [ʂɿ sɪ sɻ]*

Semibold **sī [sɿ^{55}] cí [ts^{h}ɿ^{35}] [sɿ̄ sɿ́ sɿ̌ sɿ̀] [sɻ̃] [ʂɿ sɪ sɻ]**

Semibold italic ***sī [sɿ^{55}] cí [ts^{h}ɿ^{35}] [sɿ̄ sɿ́ sɿ̌ sɿ̀] [sɻ̃] [ʂɿ sɪ sɻ]***

l below (to-do: semibold, semibold italic l below were regular's (135, −110); patched now (150, −110), (138, −110)): syllabic, voiceless, dental l, and composed (CGJ) beside precomposed l̥ l̪ l̩.

Regular [ˈbɒtl̥] [ˈlɪtl̥] [pl̥eɪ] [kl̥ɪə] [l̪̊] [l̪] [ɬ] l̥̀ ĺ̥ l̪̊ ĺ̪ l̩̊ ĺ̩ [l̥ n̥ l̪̊ ŋ̊]

Italic *[ˈbɒtl̥] [ˈlɪtl̥] [pl̥eɪ] [kl̥ɪə] [l̪̊] [l̪] [ɬ] l̥̀ ĺ̥ l̪̊ ĺ̪ l̩̊ ĺ̩ [l̥ n̥ l̪̊ ŋ̊]*

Semibold **[ˈbɒtl̥] [ˈlɪtl̥] [pl̥eɪ] [kl̥ɪə] [l̪̊] [l̪] [ɬ] l̥̀ ĺ̥ l̪̊ ĺ̪ l̩̊ ĺ̩ [l̥ n̥ l̪̊ ŋ̊]**

Semibold italic ***[ˈbɒtl̥] [ˈlɪtl̥] [pl̥eɪ] [kl̥ɪə] [l̪̊] [l̪] [ɬ] l̥̀ ĺ̥ l̪̊ ĺ̪ l̩̊ ĺ̩ [l̥ n̥ l̪̊ ŋ̊]***

ɲ above (to-do: italic ɲ above at 654, n at 645): each triple is ɲ+mark, n+CGJ+mark, precomposed n (or ŋ+mark).

Regular ɲ́ ń́ ṅ́ ɲ̀ ǹ ṅ̀ ɲ̃ ñ ṅ̃ ɲ̄ n̄ ŋ̄ ɲ̌ ň ṅ̌ ɲ̂ n̂ ŋ̂ ɲ̊ n̊ ŋ̊

Italic *ɲ́ ń ṅ́ ɲ̀ ǹ ṅ̀ ɲ̃ ñ ṅ̃ ɲ̄ n̄ ŋ̄ ɲ̌ ň ṅ̌ ɲ̂ n̂ ŋ̂ ɲ̊ n̊ ŋ̊*

Semibold **ɲ́ ń ṅ́ ɲ̀ ǹ ṅ̀ ɲ̃ ñ ṅ̃ ɲ̄ n̄ ŋ̄ ɲ̌ ň ṅ̌ ɲ̂ n̂ ŋ̂ ɲ̊ n̊ ŋ̊**

Semibold italic ***ɲ́ ń ṅ́ ɲ̀ ǹ ṅ̀ ɲ̃ ñ ṅ̃ ɲ̄ n̄ ŋ̄ ɲ̌ ň ṅ̌ ɲ̂ n̂ ŋ̂ ɲ̊ n̊ ŋ̊***

Regular Spanish año [ˈaɲo] French agneau [aɲo]

Italic *Spanish año [ˈaɲo] French agneau [aɲo]*

Semibold **Spanish año [ˈaɲo] French agneau [aɲo]**

Semibold italic ***Spanish año [ˈaɲo] French agneau [aɲo]***

Regular [ɲ̊a] [ɲ́ ɲ̀ ɲ̄] [ɲ̥] [ɲ̰] Yorùbá-like [ɲ́ ɲ̀]

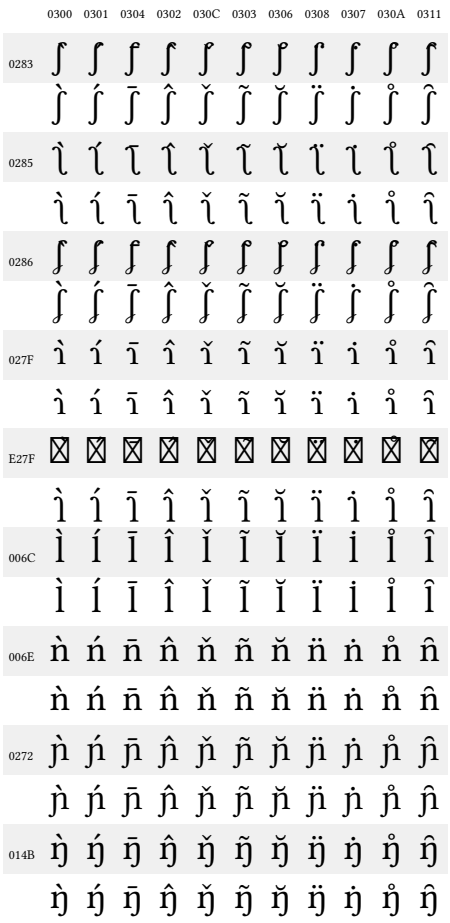
Italic *[ɲ̊a] [ɲ́ ɲ̀ ɲ̄] [ɲ̥] [ɲ̰] Yorùbá-like [ɲ́ ɲ̀]*

Semibold **[ɲ̊a] [ɲ́ ɲ̀ ɲ̄] [ɲ̥] [ɲ̰] Yorùbá-like [ɲ́ ɲ̀]**

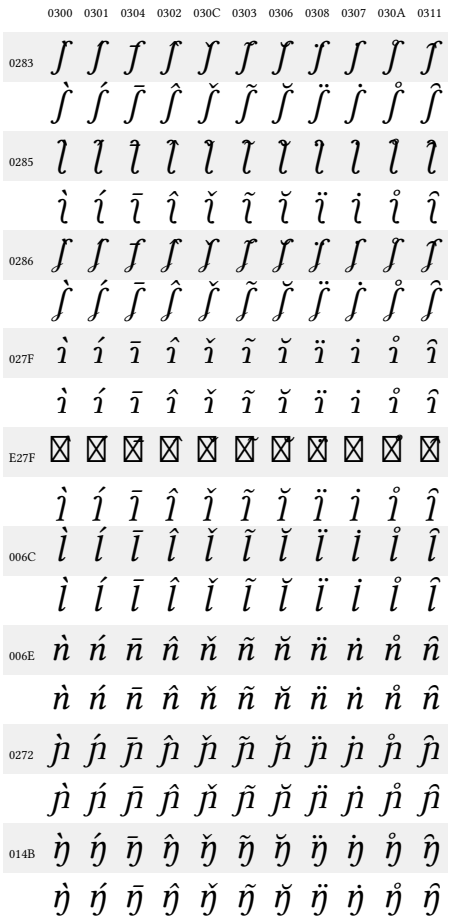
Semibold italic ***[ɲ̊a] [ɲ́ ɲ̀ ɲ̄] [ɲ̥] [ɲ̰] Yorùbá-like [ɲ́ ɲ̀]***

3. Above marks (with CGJ, so every combination uses GPOS; grey rows original, white rows patched)

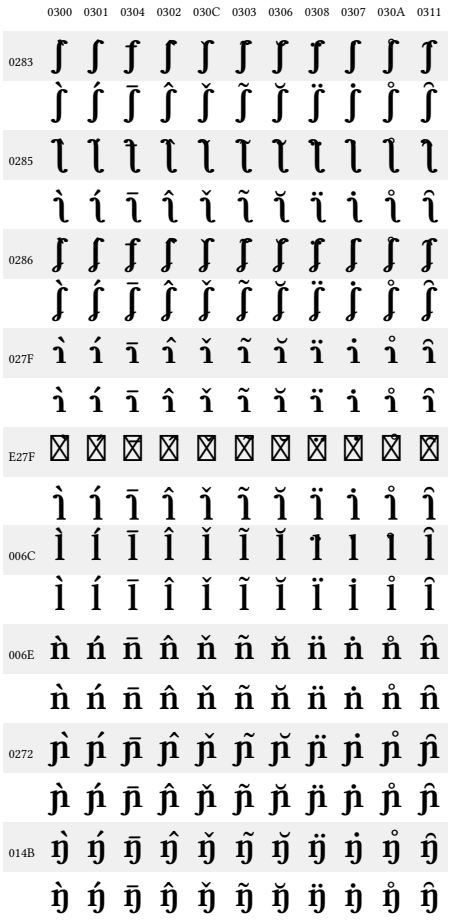
Regular



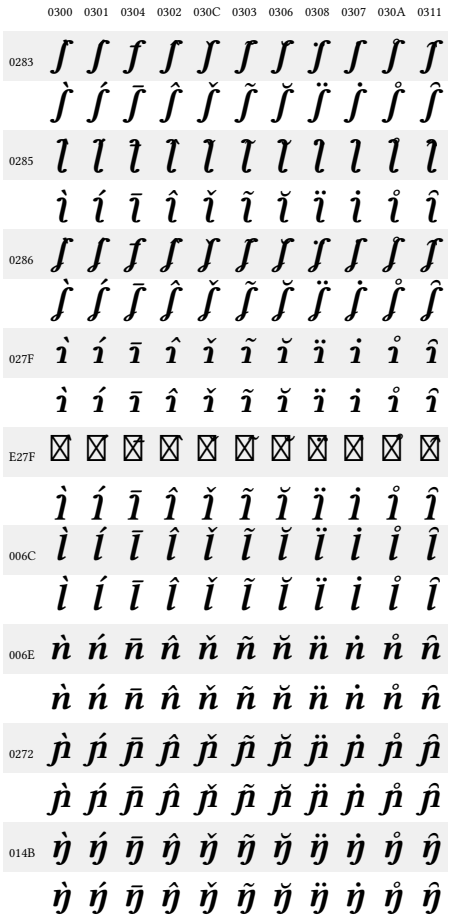
Italic



Semibold

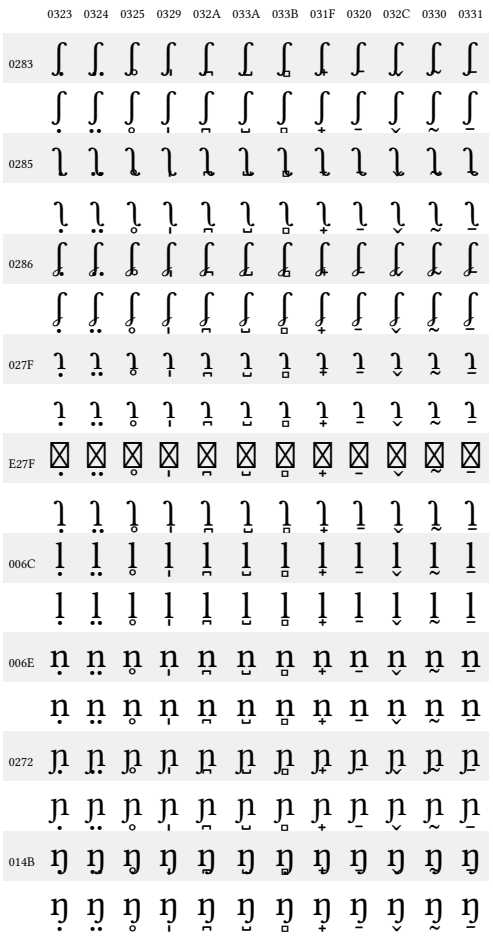


Semibold italic

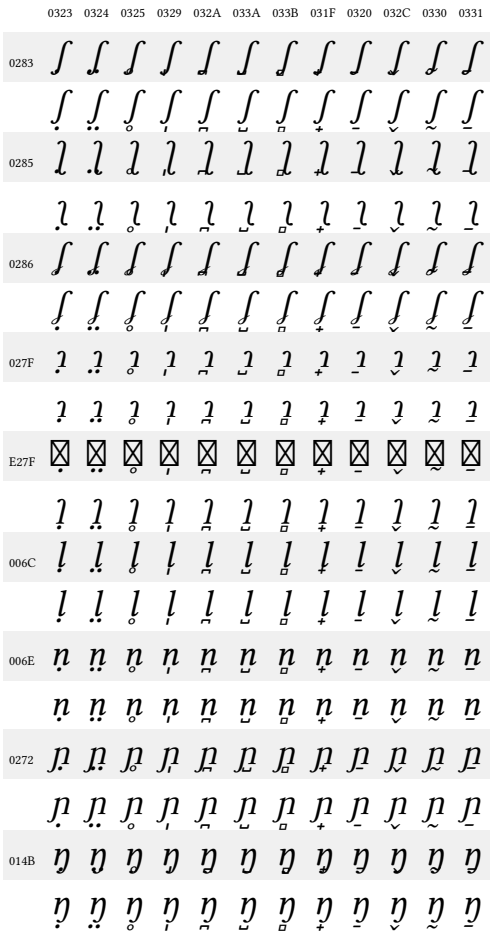


3. Below marks (with CGJ, so every combination uses GPOS; grey rows original, white rows patched)

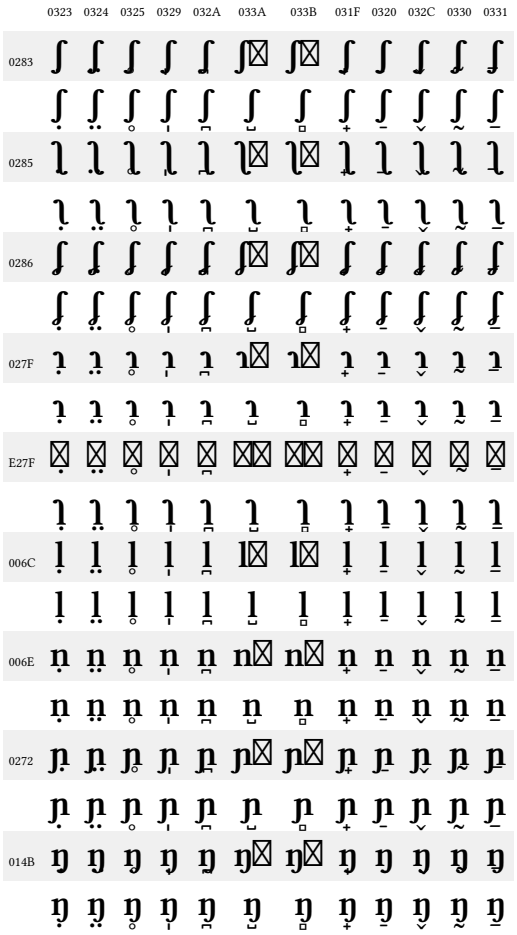
Regular



Italic



Semibold



Semibold italic

